

XpressLaundry - Final Tech Stack

Overview

XpressLaundry is a multi-vendor laundry platform consisting of:

- Customer Mobile App
- Vendor Mobile App
- Super Admin Dashboard

The platform enables customers to place laundry orders, vendors to manage laundry operations, and the platform owner to manage vendors, commissions, and payouts.

Frontend

Mobile Applications

Framework

- React Native
- Expo

Applications

- Customer App
- Vendor App

Navigation

- React Navigation

State Management

- Zustand

API Communication

- Axios

Forms

- React Hook Form

UI Components

- React Native Paper

Push Notifications

- Firebase Cloud Messaging (FCM)
-

Admin Dashboard

Framework

- React Native Web (Expo Web)

Alternative:

- Next.js (Future Upgrade)

Purpose:

- Vendor Management
 - Order Management
 - Commission Tracking
 - Payout Management
-

Backend

Framework

FastAPI

Language:

- Python 3.12

Core Packages:

```
fastapi
uvicorn
sqlalchemy
sqlmodel
pydantic
```

```
python-jose  
passlib
```

Database

Supabase PostgreSQL

Used For:

```
Users  
Vendors  
Addresses  
Orders  
Order Items  
Services  
Coupons  
Payments  
Payouts
```

Benefits:

- Managed PostgreSQL
 - Automatic Backups
 - Scalable
 - Free Tier Available
-

Authentication

Supabase Auth

Authentication Methods:

- Phone OTP Login
- Google Login

Benefits:

- No OTP Infrastructure Required
- Secure Authentication
- JWT Management Handled by Supabase

Storage

Supabase Storage

Used For:

- User Profile Photos
- Vendor Logos
- Order Images
- Invoice PDFs

Benefits:

- Integrated with Supabase
 - Simple Access Control
 - Free Tier Available
-

Payments

Razorpay

Supported Methods:

- UPI
 - Credit Cards
 - Debit Cards
 - Net Banking
-

Commission System

Weekly Payout Model

Customer Pays Online

↓

Platform Receives Payment

↓

Commission Automatically Calculated



Vendor Earnings Updated



Weekly Vendor Payout

Example

Order Amount = ₹500

Commission = 10%

Platform Commission = ₹50

Vendor Earnings = ₹450

Vendor Earnings Tracking

Vendor Dashboard

Displays:

Available Payout

₹12,450

Vendor Profile

Earnings Summary

This Week

Orders: 45

Gross Revenue: ₹22,500

Commission: ₹2,250

Net Earnings: ₹20,250

Notifications

Firestore Cloud Messaging (FCM)

Used For:

Order Placed

Pickup Scheduled

Picked Up

Washing Started

Out For Delivery

Delivered

Maps

Google Maps API

Used For:

- Address Selection
 - Pickup Locations
 - Service Coverage Areas
-

Monitoring

Sentry

Tracks:

- App Crashes
 - API Errors
 - Exceptions
-

Analytics

PostHog

Tracks:

- User Signups
 - Orders
 - Revenue
 - Customer Retention
-

Hosting

Backend Hosting

Render

Hosted Services:

- FastAPI Backend
- Background Workers

Benefits:

- Easy Deployment
 - Auto SSL
 - GitHub Integration
-

Database Hosting

Supabase

Hosted Services:

- PostgreSQL Database
 - Authentication
 - Storage
-

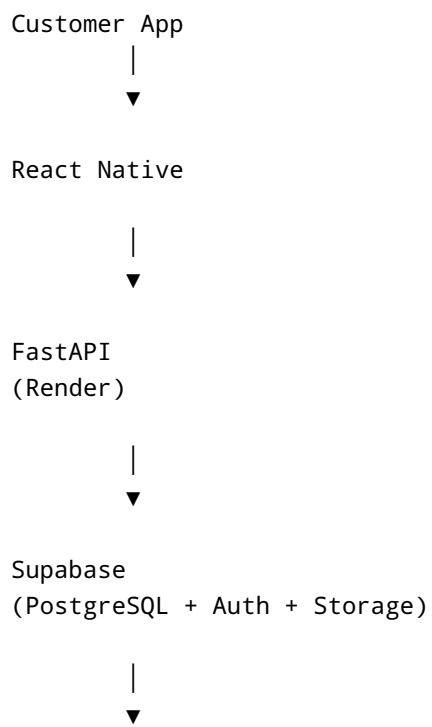
Frontend Hosting

Vercel

Used For:

- Admin Dashboard
 - Vendor Web Dashboard (Future)
-

Architecture



Razorpay



Vendor App



Super Admin Dashboard

Project Structure

backend/

```
|— app
|   |— api
|   |— models
|   |— services
|   |— repositories
|   |— auth
|   |— utils
|   └— main.py
```

frontend/

```
|— customer-app
|— vendor-app
└— admin-dashboard
```

MVP Features

Customer App

- Phone OTP Login
- Google Login
- Browse Services
- Select Laundry Package
- Add Dry Cleaning Items

- Schedule Pickup
 - Online Payment
 - Track Orders
 - Wallet & Coupons
 - Profile Management
-

Vendor App

- Dashboard
 - Orders Management
 - Customer Management
 - Services Management
 - Earnings Tracking
 - Available Payout Tracking
 - Profile Management
-

Super Admin Dashboard

- Vendor Management
 - Orders Monitoring
 - Revenue Tracking
 - Commission Tracking
 - Payout Management
 - Commission Configuration
-

Estimated Monthly Cost

Supabase	₹0
Render	₹0 - ₹600
Firebase FCM	₹0
PostHog	₹0
Sentry	₹0

Total Estimated Cost

₹500 - ₹1000 / month

Final Stack Summary

Frontend:

- React Native
- Expo
- React Navigation
- Zustand
- Axios

Backend:

- FastAPI
- Python

Database:

- Supabase PostgreSQL

Authentication:

- Supabase Auth

Storage:

- Supabase Storage

Payments:

- Razorpay

Notifications:

- Firebase Cloud Messaging

Hosting:

- Render
- Supabase

Monitoring:

- Sentry

Analytics:

- PostHog